UNIVERSITY: _____

COLLEGE: _____

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

WEB TECHNOLOGY
LABORATORY FILE

Student Name: _____

Roll Number: _____

Enrollment Number: _____

Course: B.Tech CSE

Semester: _____

Academic Session: _____

Submitted To (Faculty Name): _____

INDEX

S. No. / Name of Program	Date	Page No.
1. HTML program to create resume preparation using tables.		3
2. HTML program for home page creation using frames.		4
3. HTML program for form creation.		5
4. Create a web page to embed an image map in a web page using HTML.		6
5. Create a web page to fix the hot spots and to show all the related information when the hot spots are clicked using HTML.		7
6. Create a web page to get the coordinates from an image using JavaScript.		8
7. Create a web page with all types of cascading style sheets.		9
8. Write HTML/JavaScript to display your CV in navigator, your institute website, Department website and tutorial website for specific subject.		10
9. Design HTML form for keeping student record and validate it using JavaScript.		12
10. Writing program in XML for creation of DTD, which specifies set of rules.		13
11. Create a style sheet in CSS/XSL and display the document in Internet Explorer.		14

Practical No. 1

Program: HTML program to create resume preparation using tables.

Source Code:

resume.html

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><title>Resume</title></head>
<body>
  <h1>Resume</h1>
  <table border="1" cellpadding="8">
    <tr><th>Name</th><td>Your Name</td></tr>
    <tr><th>Email</th><td>you@example.com</td></tr>
    <tr><th>Education</th><td>B.Tech in Computer Science</td></tr>
    <tr><th>Skills</th><td>HTML, CSS and JavaScript</td></tr>
    <tr><th>Interests</th><td>Web development</td></tr>
  </table>
</body>
</html>
```

Output/Result:

The browser displays a Resume heading and a bordered table containing sample personal, education and skill details.

Practical No. 2

Program: HTML program for home page creation using frames.

Source Code:

index.html

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Frameset//EN"
  "http://www.w3.org/TR/html4/frameset.dtd">
<html>
<head><title>Frames Home Page</title></head>
<frameset cols="25%,75%">
  <frame src="menu.html" title="Menu">
  <frame src="home.html" name="content" title="Content">
  <noframes><body>Frames are not supported.</body></noframes>
</frameset>
</html>
```

menu.html

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><title>Menu</title></head>
<body>
  <h2>Menu</h2>
  <p><a href="home.html" target="content">Home</a></p>
  <p><a href="about.html" target="content">About</a></p>
</body>
</html>
```

home.html

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><title>Home</title></head>
<body><h1>Welcome</h1><p>This is the home page.</p></body>
</html>
```

about.html

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><title>About</title></head>
<body><h1>About</h1><p>This is a sample frames page.</p></body>
</html>
```

Output/Result:

Open index.html. A menu appears on the left and the home page on the right. Clicking About changes the right frame. Frames are obsolete in HTML5 and may not work in all modern environments.

Practical No. 3

Program: HTML program for form creation.

Source Code:

form.html

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><title>Registration Form</title></head>
<body>
  <h1>Registration Form</h1>
  <form action="#" method="get">
    <p><label>Name: <input type="text" name="name"></label></p>
    <p><label>Email: <input type="email" name="email"></label></p>
    <p>Gender:
      <label><input type="radio" name="gender" value="male"> Male</label>
      <label><input type="radio" name="gender" value="female"> Female</label>
    </p>
    <p><label>Course:
      <select name="course"><option>B.Tech CSE</option>
        <option>B.Tech IT</option></select></label></p>
    <p><label>Address: <textarea name="address"></textarea></label></p>
    <button type="submit">Submit</button>
    <button type="reset">Reset</button>
  </form>
</body>
</html>
```

Output/Result:

A registration form appears with name, email, gender, course and address fields, along with Submit and Reset buttons.

Practical No. 4

Program: Create a web page to embed an image map in a web page using HTML.

Source Code:

image-map.html

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><title>Image Map</title></head>
<body>
  <h1>Image Map</h1>
  <p>Click a shape to visit its information page.</p>
  <img src="shapes.svg" width="400" height="200"
       alt="A blue rectangle and an orange circle" usemap="#shapes">
  <map name="shapes">
    <area shape="rect" coords="30,40,150,140" alt="Rectangle"
          href="https://en.wikipedia.org/wiki/Rectangle">
    <area shape="circle" coords="270,90,50" alt="Circle"
          href="https://en.wikipedia.org/wiki/Circle">
  </map>
</body>
</html>
```

shapes.svg

```
<svg xmlns="http://www.w3.org/2000/svg" width="400" height="200">
  <rect width="400" height="200" fill="#f2f2f2"/>
  <rect x="30" y="40" width="120" height="100" fill="steelblue"/>
  <circle cx="270" cy="90" r="50" fill="orange"/>
</svg>
```

Output/Result:

The page displays a blue rectangle and an orange circle. Clicking either shape opens the corresponding information page; internet access is required.

Practical No. 5

Program: Create a web page to fix the hot spots and to show all the related information when the hot spots are clicked using HTML.

Source Code:

hotspots.html

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><title>Shape Hot Spots</title></head>
<body>
  <h1>Shape Hot Spots</h1>
  <p>Click a shape to see its information below.</p>
  <img src="shapes.svg" width="400" height="200"
       alt="A blue rectangle and an orange circle" usemap="#shapes">
  <map name="shapes">
    <area shape="rect" coords="30,40,150,140" alt="Rectangle"
          href="#rectangle">
    <area shape="circle" coords="270,90,50" alt="Circle"
          href="#circle">
  </map>
  <h2 id="rectangle">Rectangle</h2>
  <p>A rectangle has four sides and four right angles.</p>
  <h2 id="circle">Circle</h2>
  <p>A circle consists of points equally distant from its centre.</p>
</body>
</html>
```

shapes.svg

```
<svg xmlns="http://www.w3.org/2000/svg" width="400" height="200">
  <rect width="400" height="200" fill="#f2f2f2"/>
  <rect x="30" y="40" width="120" height="100" fill="steelblue"/>
  <circle cx="270" cy="90" r="50" fill="orange"/>
</svg>
```

Output/Result:

The image contains two clickable hot spots. Clicking the rectangle or circle moves to the related information on the same page.

Practical No. 6

Program: Create a web page to get the coordinates from an image using JavaScript.

Source Code:

coordinates.html

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><title>Image Coordinates</title></head>
<body>
  <h1>Image Coordinates</h1>
  <p>Click anywhere on the image.</p>
  <img id="picture" src="shapes.svg" width="400" height="200"
       alt="A blue rectangle and an orange circle">
  <p id="result">Coordinates: not selected</p>
  <script>
    const picture = document.getElementById('picture');
    picture.addEventListener('click', function (event) {
      const box = picture.getBoundingClientRect();
      const x = Math.floor((event.clientX - box.left) *
                           picture.naturalWidth / box.width);
      const y = Math.floor((event.clientY - box.top) *
                           picture.naturalHeight / box.height);
      document.getElementById('result').textContent =
        'Coordinates: X = ' + x + ', Y = ' + y;
    });
  </script>
</body>
</html>
```

shapes.svg

```
<svg xmlns="http://www.w3.org/2000/svg" width="400" height="200">
  <rect width="400" height="200" fill="#f2f2f2"/>
  <rect x="30" y="40" width="120" height="100" fill="steelblue"/>
  <circle cx="270" cy="90" r="50" fill="orange"/>
</svg>
```

Output/Result:

Clicking the image displays the X and Y coordinates in image pixels, measured from the top-left corner. For example, clicking near the circle centre gives approximately X = 270, Y = 90.

Practical No. 7

Program: Create a web page with all types of cascading style sheets.

Source Code:

styles.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8"><title>Types of CSS</title>
  <link rel="stylesheet" href="style.css">
  <style>
    .internal { color: green; }
  </style>
</head>
<body>
  <h1>Types of CSS</h1>
  <p style="color: red;">This text uses inline CSS.</p>
  <p class="internal">This text uses internal CSS.</p>
  <p class="external">This text uses external CSS.</p>
</body>
</html>
```

style.css

```
.external {
  color: blue;
  font-weight: bold;
}
```

Output/Result:

The page displays three statements: inline CSS in red, internal CSS in green and external CSS in bold blue.

Practical No. 8

Program: Write HTML/JavaScript to display your CV in navigator, your institute website, Department website and tutorial website for specific subject.

Source Code:

navigator.html

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><title>Web Navigator</title></head>
<body>
  <h1>Web Navigator</h1>
  <p>Choose a page to open in a new tab:</p>
  <ul>
    <li><a href="cv.html" target="_blank">My CV</a></li>
    <li><a id="institute" href="#">Institute website</a></li>
    <li><a id="department" href="#">Department website</a></li>
    <li><a href="https://developer.mozilla.org/en-US/docs/Web/JavaScript"
           target="_blank" rel="noopener">JavaScript tutorial</a></li>
  </ul>
  <script>
    // Replace these empty strings with your actual website URLs.
    const instituteURL = '';
    const departmentURL = '';
    function openSite(url) {
      if (url) window.open(url, '_blank', 'noopener');
      else alert('Please add the website URL in the code.');
    }
    document.getElementById('institute').onclick = function (e) {
      e.preventDefault(); openSite(instituteURL);
    };
    document.getElementById('department').onclick = function (e) {
      e.preventDefault(); openSite(departmentURL);
    };
  </script>
</body>
</html>
```

Practical No. 8 (continued)

Source Code (continued):

cv.html

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><title>My CV</title></head>
<body>
  <h1>My CV</h1>
  <p>Name: Your Name</p>
  <p>Course: B.Tech CSE</p>
  <p>Skills: HTML, CSS and JavaScript</p>
</body>
</html>
```

Output/Result:

The navigator lists links to a sample CV, institute, department and a JavaScript tutorial. Enter the actual institute and department URLs in the two variables before using those links; otherwise an alert appears. External sites require internet access.

Practical No. 9

Program: Design HTML form for keeping student record and validate it using JavaScript.

Source Code:

student-record.html

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><title>Student Record</title></head>
<body>
  <h1>Student Record</h1>
  <form id="studentForm" novalidate>
    <p><label>Name: <input id="name" type="text"></label></p>
    <p><label>Roll number: <input id="roll" type="text"></label></p>
    <p><label>Email: <input id="email" type="email"></label></p>
    <p><label>Age: <input id="age" type="number"></label></p>
    <button type="submit">Save Record</button>
  </form>
  <p id="message" aria-live="polite"></p>
  <script>
    document.getElementById('studentForm').onsubmit = function (e) {
      e.preventDefault();
      const name = document.getElementById('name').value.trim();
      const roll = document.getElementById('roll').value.trim();
      const email = document.getElementById('email');
      const age = Number(document.getElementById('age').value);
      const message = document.getElementById('message');
      if (!name || !roll || !email.value || !email.checkValidity() ||
          !Number.isInteger(age) || age < 16 || age > 100) {
        message.textContent = 'Enter all fields and a valid email; ' +
                              'age must be 16 to 100.';
        return;
      }
      const record = { name: name, roll: roll,
                       email: email.value, age: age };
      localStorage.setItem('studentRecord', JSON.stringify(record));
      message.textContent = 'Student record saved on this browser.';
    };
  </script>
</body>
</html>
```

Output/Result:

Invalid or empty inputs show an error. Valid details show a success message and save one student record in the browser's local storage (a later save replaces it).

Practical No. 10

Program: Writing program in XML for creation of DTD, which specifies set of rules.

Source Code:

student.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE student [
  <!ELEMENT student (name, roll, course)>
  <!ELEMENT name (#PCDATA)>
  <!ELEMENT roll (#PCDATA)>
  <!ELEMENT course (#PCDATA)>
]>
<student>
  <name>Your Name</name>
  <roll>101</roll>
  <course>B.Tech CSE</course>
</student>
```

Output/Result:

The browser displays the XML document as a tree with student, name, roll and course elements. The internal DTD specifies their required order; use a DTD validator to check validity, as browsers may not validate it.

Practical No. 11

Program: Create a style sheet in CSS/XSL and display the document in Internet Explorer.

Source Code:

students.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="students.xsl"?>
<students>
  <student><name>Student One</name><course>B.Tech CSE</course></student>
  <student><name>Student Two</name><course>B.Tech CSE</course></student>
</students>
```

students.xsl

```
<?xml version="1.0" encoding="UTF-8"?>
<xsl:stylesheet version="1.0"
  xmlns:xsl="http://www.w3.org/1999/XSL/Transform">
  <xsl:template match="/">
    <html>
      <head><title>Student List</title>
        <style>table, th, td { border: 1px solid black; }
               table { border-collapse: collapse; }</style>
      </head>
      <body>
        <h1>Student List</h1>
        <table>
          <tr><th>Name</th><th>Course</th></tr>
          <xsl:for-each select="students/student">
            <tr><td><xsl:value-of select="name"/></td>
                <td><xsl:value-of select="course"/></td></tr>
          </xsl:for-each>
        </table>
      </body>
    </html>
  </xsl:template>
</xsl:stylesheet>
```

Output/Result:

Open students.xml to see a bordered Student List table with two rows. Internet Explorer 11 supports XSLT 1.0 but is retired; Edge IE mode may be used where available. In modern browsers, serve both files from the same folder over HTTP (for example, python -m http.server) and open the XML URL; local-file XSL access may be blocked.